In [ ]:

```
import cv2
import matplotlib.pyplot as plt
import numpy as np

# STEP 1: Import Required Libraries & Setup  // Pragya Singhal , CU24220050 , 43

print("Libraries imported successfully!")
print(f"OpenCV Version: {cv2.__version__}")
print(f"NumPy Version: {np.__version__}")


# STEP 2: Load a Color Image
# Replace 'sample_image.jpg' with the path to your image file
image_path = "sample_image.jpg"

# Note: OpenCV loads images in BGR format by default
img_bgr = cv2.imread(image_path)

if img_bgr is None:
  print(
      f"Error: Could not load image from '{image_path}'. Please check the"
      " file path."
  )
else:
  # 2. Display using OpenCV (Converts BGR to RGB for correct colors in standard vie
  # cv2.imshow("OpenCV Window", img_bgr)
  # cv2.waitKey(0)
  # cv2.destroyAllWindows()

  # Convert BGR to RGB for Matplotlib visualization
  img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

  plt.figure(figsize=(6, 6))
  plt.imshow(img_rgb)
  plt.title("Original Image (Displayed via Matplotlib)")
  plt.axis("off")
  plt.show()

  # STEP 3: Examine Image Properties    // Pragya Singhal , CU24220050 , 43
  print("\n--- Image Properties ---")
  print(f"Dimensions (Height, Width, Channels): {img_bgr.shape}")
  print(f"Total number of pixels: {img_bgr.size}")
  print(f"Data type of pixels: {img_bgr.dtype}")

  #
  # STEP 4: Save Image in Different Formats   // Pragya Singhal , CU24220050 , 43
  #
  cv2.imwrite("output_saved.jpg", img_bgr, [cv2.IMWRITE_JPEG_QUALITY, 90])
  cv2.imwrite("output_saved.png", img_bgr, [cv2.IMWRITE_PNG_COMPRESSION, 3])
  print(
      "\nImages saved successfully as 'output_saved.jpg' (JPEG) and"
      " 'output_saved.png' (PNG)."
  )
```

```
#
# STEP 5: Color Space Conversions   // Pragya Singhal , CU24220050 , 43
#
gray = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2GRAY)
hsv = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2HSV)
lab = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2LAB)

# Plotting color spaces
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale Space")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(hsv)
plt.title("HSV Space")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(lab)
plt.title("LAB Space")
plt.axis("off")
plt.tight_layout()
plt.show()

# STEP 6: Geometric Transformations  // Pragya Singhal , CU24220050 , 43

height, width = img_bgr.shape[:2]

# Resize (scaling to 50% of original dimensions)
resized = cv2.resize(img_bgr, (width // 2, height // 2), interpolation=cv2.INTER_

# Rotation (Rotate 45 degrees around center)
center = (width // 2, height // 2)
matrix = cv2.getRotationMatrix2D(center, 45, 1.0)
rotated = cv2.warpAffine(img_bgr, matrix, (width, height))

# Flipping (Horizontal: 1, Vertical: 0)  // Pragya Singhal , CU24220050 , 43
flipped_horizontal = cv2.flip(img_bgr, 1)
flipped_vertical = cv2.flip(img_bgr, 0)

# STEP 7: Generate Image Complement (Negative)
# For 8-bit images, the negative is obtained by subtracting pixel values from 255
negative_img = 255 - img_bgr

#
# STEP 8: Crop Region of Interest (ROI)

# Slicing array [startY:endY, startX:endX]
# Let's crop a central region
start_y, end_y = int(height * 0.25), int(height * 0.75)
start_x, end_x = int(width * 0.25), int(width * 0.75)
roi = img_bgr[start_y:end_y, start_x:end_x]
roi_rgb = cv2.cvtColor(roi, cv2.COLOR_BGR2GRAY)  # Example: converting ROI to gra
```

```python
# STEP 9: Comprehensive Comparison Display   // Pragya Singhal , CU24220050 , 43
operations = [
    ("Original RGB", img_rgb),
    ("Grayscale", cv2.cvtColor(gray, cv2.COLOR_GRAY2RGB)),
    ("Negative Image", cv2.cvtColor(negative_img, cv2.COLOR_BGR2RGB)),
    ("Rotated (45°)", cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB)),
    ("Flipped (Horizontal)", cv2.cvtColor(flipped_horizontal, cv2.COLOR_BGR2RGB))
    ("Cropped ROI", cv2.cvtColor(roi, cv2.COLOR_BGR2RGB)),
]

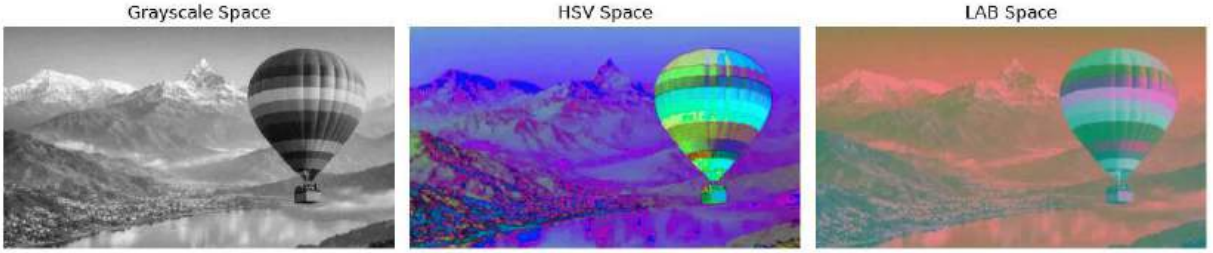
plt.figure(figsize=(12, 8))
for i, (title, img) in enumerate(operations):
  plt.subplot(2, 3, i + 1)
  plt.imshow(img)
  plt.title(title)
  plt.axis("off")
plt.tight_layout()
plt.show()
```

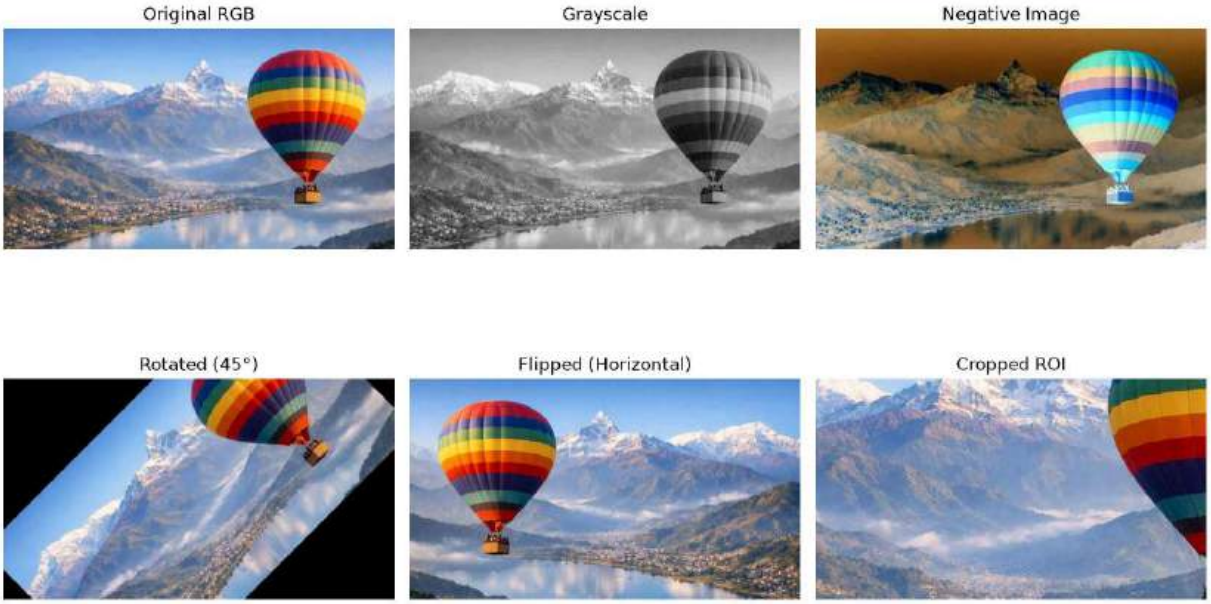
```
Libraries imported successfully!
OpenCV Version: 5.0.0
NumPy Version: 2.5.2
```




```
--- Image Properties ---
Dimensions (Height, Width, Channels): (675, 1200, 3)
Total number of pixels: 2430000
Data type of pixels: uint8

Images saved successfully as 'output_saved.jpg' (JPEG) and 'output_saved.png' (PNG).
```




Original RGB
Grayscale
Negative Image
Rotated (45°)
Flipped (Horizontal)
Cropped ROI